REPORT File OF CO-OP Research Paper AT INDUSTRY (MODULE-1)

ON

Mental Health Assessment Using Sentiment Analysis

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

We, Simranpreet Kaur (2210990854) and Anish Singal (2210990116), B.E.-2022 of the Chitkara University, Punjab hereby declare that the Research Paper Report entitled "**Mental Health Assessment using Sentiment Analysis**" is an original work and data provided in the study is authentic to the best of our knowledge. This report has not been submitted to any other Institute for the award of any other course.

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CANDIDATE'S DECLARATION

We, Simranpreet Kaur(2210990854) and Anish Singal(2210990116), B.E.-2022 of the Chitkara University, Punjab hereby declare that the Research Paper Report entitled "**Mental Health Assessment using Sentiment Analysis**" is an original work and data provided in the study is authentic to the best of our knowledge. This report has not been submitted to any other Institute for the award of any other course.

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(Annexure -E)

ACKNOWLEDGEMENT

It is our pleasure to be indebted to various people, who directly or indirectly contributed in the development of this work and who influenced my thinking, behavior and acts during the course of study.

We express our sincere gratitude to all for providing me an opportunity to undergo Research Paper as the part of the curriculum.

We are thankful to our Project Guide "**Dr Shikha Tuteja**" for her support, cooperation, and motivation provided to us. Her constant inspiration, valuable suggestions, and precious time played a critical role in accomplishing our research paper report.

Lastly, We would like to thank the almighty and our parents for their moral support and friends with whom we shared our day-to day experience and received lots of suggestions that improve our quality of work.

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Abstract-This research paper addresses a major worldwide concern: the rising incidence of mental health illnesses, which is frequently made worse by the lack of clinical services and the stigma associated with conventional diagnosis. The system leverages social media as a "digital footprint" for psychological evaluation in order to meet the pressing need for scalable, non-invasive screening methods. To automatically detect distress indicators in unstructured text, we provide an advanced sentiment analysis method. Our technique uses a comprehensive NLP pipeline on a large-scale dataset of 52,681 records and utilizes three classifiers: Linear SVM, Random Forest, and Multinomial Naïve Bayes. The framework successfully captured multi-dimensional diagnostic markers, distinguishing between 7 overlapping psychological categories. Linear SVM yielded optimal results, producing a balanced F1 score of 0.76 and a classification accuracy of 76.42%..

1. Introduction to the project

a. **Background** - The increasing prevalence of mental disorders such as depression, anxiety, and bipolar disorder has created a crisis in global health, putting a huge burden on the public healthcare system. Traditionally, mental health issues are diagnosed through clinical assessments or self-reported questionnaires. The digital revolution, however, has provided new ways to document human emotions. Social media sites have created spaces where people freely express their emotions regarding mental health more openly than in formal settings.

b. **Problem Statement** - Traditional diagnosis is fraught with self-stigma, financial barriers, and geographical limitations, preventing timely help. Additionally, current computational models suffer from the "Binary Constraint," overly simplifying diagnoses into merely "Depressed" or "Normal." These fail to accommodate the multidimensionality of medical language and overlapping linguistic cues (e.g., chronic stress vs. clinical anxiety). The project aims to solve this by providing an accurate, multi-class

computational infrastructure to analyze vast digital footprints for early, non-invasive psychiatric distress detection.

2. Software Requirement Specification

a. **Methods** - The system is developed using an advanced Natural Language Processing (NLP) pipeline integrated with robust machine learning architectures. The methodology involves robust noise reduction, morphological normalization via WordNet Lemmatization, and feature engineering utilizing TF-IDF Vectorization mapped to a highly dimensional sparse matrix.

b. **Programming/Working Environment**

Programming Language: Python 3.9.12
Machine Learning Libraries: Scikit Learn(v1.0.2)
Machine Learning Models: Multinomial Naive Bayes (MNB), Random Forest (RF), and Linear Support Vector Machine (SVM)
NLP libraries: NLTK(v3.7)
Operating System: Windows/Linux

3. System Architecture And Workflows:

The proposed framework for multi-class mental health screening operates sequentially:

- **Data Acquisition**: Ingestion of 52,681 raw social media records.
- **Categorization**: Labeling into 7 classes (Normal, Depression, Anxiety, Suicidal, Bipolar, Stress, Personality).
- **NLP Preprocessing**: Syntax correction, Noise Reduction, Tokenization, Stop-word Removal, and WordNet Lemmatization.
- **Feature Engineering**: TF-IDF Vectorization generating a sparse matrix capped at the top 5,000 features to capture vital sentiment-heavy lexical content.

- **Model Evaluation**: Applying Naive Bayes (MNB), Random Forest (RF), and the optimized engine, Linear SVM.

4. **Database and Dataset Analysis:**
   a. **Data Source:** To establish a robust classifier, a varied and extensive dataset consisting of 52,681 anonymized textual records was gathered from platforms like Reddit and X (formerly Twitter). Due to the "online disinhibition effect," users generate honest digital footprints necessary to map complex psychiatric profiles.
   b. **Multi-Class Categorization:** Moving away from earlier binary models, the database was organized into seven distinct categories to set mathematical bounds between overlapping states. The dataset was split utilizing an 80-20 ratio: 42,144 records for training the predictive engines and 10,537 unseen records reserved strictly for testing and validation.

5. **Code-Implementation and Database Connections:**

5.1 NLP Preprocessing Code Implementation: Raw digital communications contain severe noise. The implementation strictly strips website links, HTML tags, and numeric values before standardizing cases. Semantic units are processed by filtering non-contributing terminologies (via NLTK stop-words), followed by WordNet Lemmatization, which ensures context-aware reduction of morphological variants without compromising vital medical definitions.

5.2 Feature Extraction via TF-IDF: Machine models require numerical representations. Instead of mere frequency counts, the system implements Term Frequency-Inverse Document Frequency (TF-IDF). This mathematical weighting diminishes the importance of ubiquitous terms while emphasizing rare, diagnostically critical markers (e.g., "insomnia" or "euphoria").

5.3 Classifier Integration: Three models were coded and compared. Multinomial Naive Bayes (MNB) established a baseline (65.35% accuracy) but suffered from feature-independence flaws. Random Forest (RF) achieved 71.20% accuracy but plateaued due to the sparse dimensionality of the TF-IDF matrix. The Linear Support Vector Machine (SVM)

outperformed all, utilizing geometric hyperplane optimization in the high-dimensional space to distinguish highly analogous clinical conditions, resulting in an accuracy of 76.42%.

6. **Limitations:**

   Lack of Temporal Analysis: The current system analyzes isolated text records and does not track the chronological progression of a user's mental health state over time.

   "Stress" Classification: The model yielded a lower F1-score (0.55) specifically for the "Stress" category. Stress highly overlaps in lexical usage with generic anxiety and depression, challenging clear geometric separation.

   Computational Overhead: Processing the 5,000-feature sparse matrix requires significant hardware overhead, potentially limiting ultra-lightweight deployments.

7. **Conclusion:** The sentiment analysis system successfully delivered a scalable, non-invasive diagnostic support framework. By overcoming the standard "binary constraint," the multi-class categorization enabled the distinction of 7 psychological states from 52,681 records. Utilizing tailored NLP preprocessing and TF-IDF vectorization, the Linear Support Vector Machine emerged as the optimal classifier. Achieving a 76.42% accuracy and balanced F1-Score of 0.76, the system demonstrated significant precision in identifying high-risk states, mitigating false alarms for normal scenarios, and proving the feasibility of bridging digital informal interactions with clinical decision-making.

8. **Future Scope:** The future scope of this research includes deploying advanced deep learning architectures capable of capturing deeper contextual relationships within sentences. Furthermore, adapting the system for temporal analysis will allow tracking a user's emotional trajectory across time, determining if psychological conditions are deteriorating or improving. Expanding the models to accommodate

multimodal data (e.g., incorporating voice and image sentiment) will transform this framework into an exponentially more robust, real-time psychiatric intervention tool.

9. Bibliography/Reference:

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